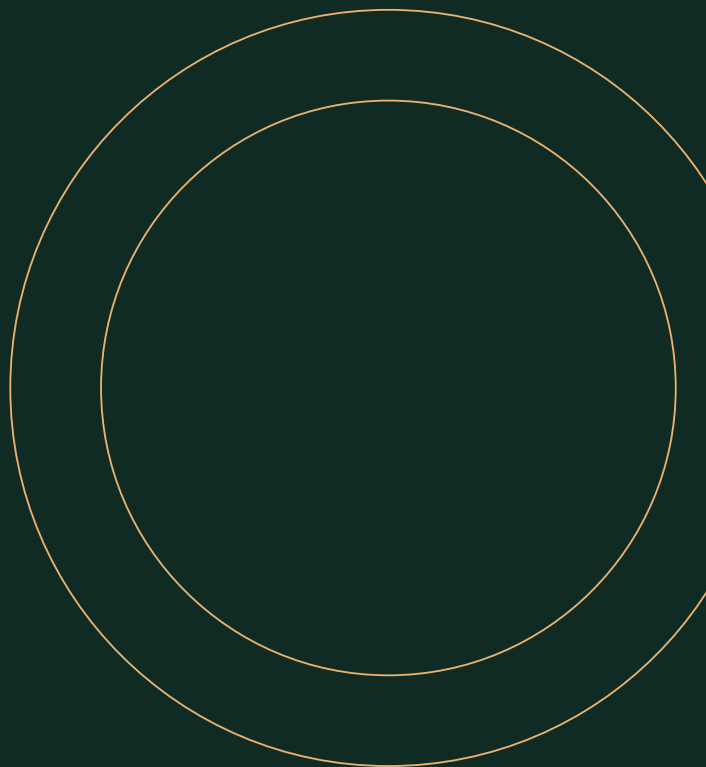


Evidence controlled.

Intelligence governed.

End-to-end product, system, data, AI, operations and deployment specification for governed commissioning intelligence.



DOCUMENT INDEX

A4 technical dossier

This print-safe edition consolidates the current repository architecture, product requirements, technical requirements, data schema, application flow, design decisions, retrieval safeguards, release ledger and browser acceptance evidence.

Part	Content
01	Purpose, target users and authority model
02	Application experience and live screenshots
03	System architecture and service boundaries
04	End-to-end data flow, user flow and navigation
05	Database, provenance and schema domains
06	AI, RAG, agents and safety controls
07	Technology stack, SDKs, APIs and third parties
08	Prerequisites, installation, environment and operations
09	Testing strategy, branch/application errors and mitigations
10	Deployment readiness and future CI/CD

01 / PURPOSE, USERS AND AUTHORITY

What Pramana Cx is designed to decide

Pramana Cx is a governed evidence control plane for mission-critical EPC and data-centre commissioning. It connects controlled requirements to systems, assets, gates, tests, evidence, findings, schedules, shipments and immutable source evidence so an authorized engineer can make a defensible gate decision.

Non-negotiable authority rule. AI may extract, retrieve, rank, draft, classify and explain. Deterministic services calculate readiness, schedule feasibility and structured verdicts. Only an authorized human can accept a requirement or evidence item, approve a report, disposition a compliance finding or make a gate decision.

Target users and friction removed

User	Decision	How the application makes it easier
Commissioning manager / authority	Can a gate be defended today?	Readiness unifies accepted proof, stale/failed evidence, predecessors, blockers, owners and citations.
EPC / GC MEP package lead	What must be fixed before the next test?	Actions, Cx, evidence and schedule views preserve shared operational context.
QA/QC engineer	Does a controlled record meet a requirement?	Source-region provenance, deterministic checks and cited comparison proposals replace evidence hunting.
Planner / scheduler	What changed and should the baseline move?	Reviewed inputs, immutable CP-SAT versions and before/after explanations separate facts from speculation.
Owner / operations team	Is turnover complete and auditable?	Approved-gate-only manifests, hashes and independent verification make the package inspectable.
Field engineer / vendor	How can proof be captured safely on site?	Offline capture queues evidence idempotently until an authorized reviewer accepts it.

Application coverage

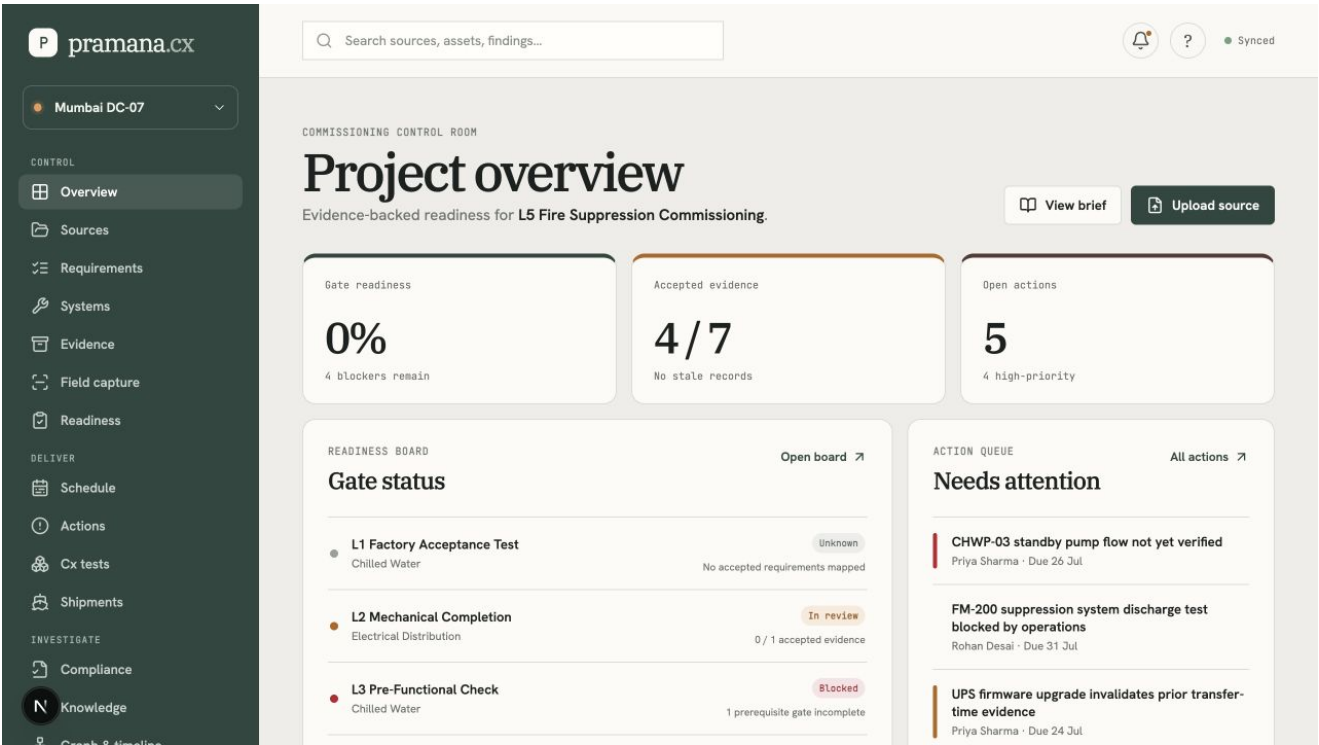
- Controlled sources, revisions, exact source-region citations and proposal review.
- Requirements, systems, assets, gates, evidence, findings, readiness and turnover packs.
- Schedule inputs, CP-SAT baselines, immutable versions, live events and advisory risks.
- Cx standards, cited checklists, deterministic readings, reports and human approval.
- Shipments, map routes, compliance, knowledge/RFI intelligence, graph, changes and Command Center.

02 / APPLICATION EXPERIENCE

Task-oriented, cited and resilient

The workspace groups navigation as Control, Deliver and Investigate. Deep links preserve the selected gate, task, finding or shipment across feature boundaries. Busy-state and progress behaviour is operation-specific, bounded and visible, so long-running work cannot silently replace another action.

Live browser-captured application surfaces



Overview and grouped navigation - one project context and a complete task map.

P

pramana.cx

Mumbai DC-07

CONTROL

Overview

Sources

Requirements

Systems

Evidence

Field capture

Readiness

DELIVER

Schedule

Actions

Cx tests

Shipments

INVESTIGATE

Compliance

N Knowledge

Create & timeline

Search sources, assets, findings...

OnLine · server connected

COMMISSIONING QA COPILOT

Cx tests

Ingest controlled standards, generate citation-verified advisory checklists, execute deterministic proposed checks, and approve an editable report into immutable evidence.

Overview

1 · CONTROLLED CORPUS

Ingest standards/procedures

Checklist generation remains disabled until extraction creates exact citation regions.

STANDARD SET

ASHRAE / project IST set

TITLE

Synthetic chilled-water IS

DOCUMENT TYPE

Standard

REVISION

Rev 1

PDF

Choose file

No file chosen

Control & extract

2 · ADVISORY GENERATION

Generate cited draft

5 extracted standard version(s) available.

TITLE

L4 integrated systems test

SYSTEM

Select system

GATE

Select gate

EQUIPMENT

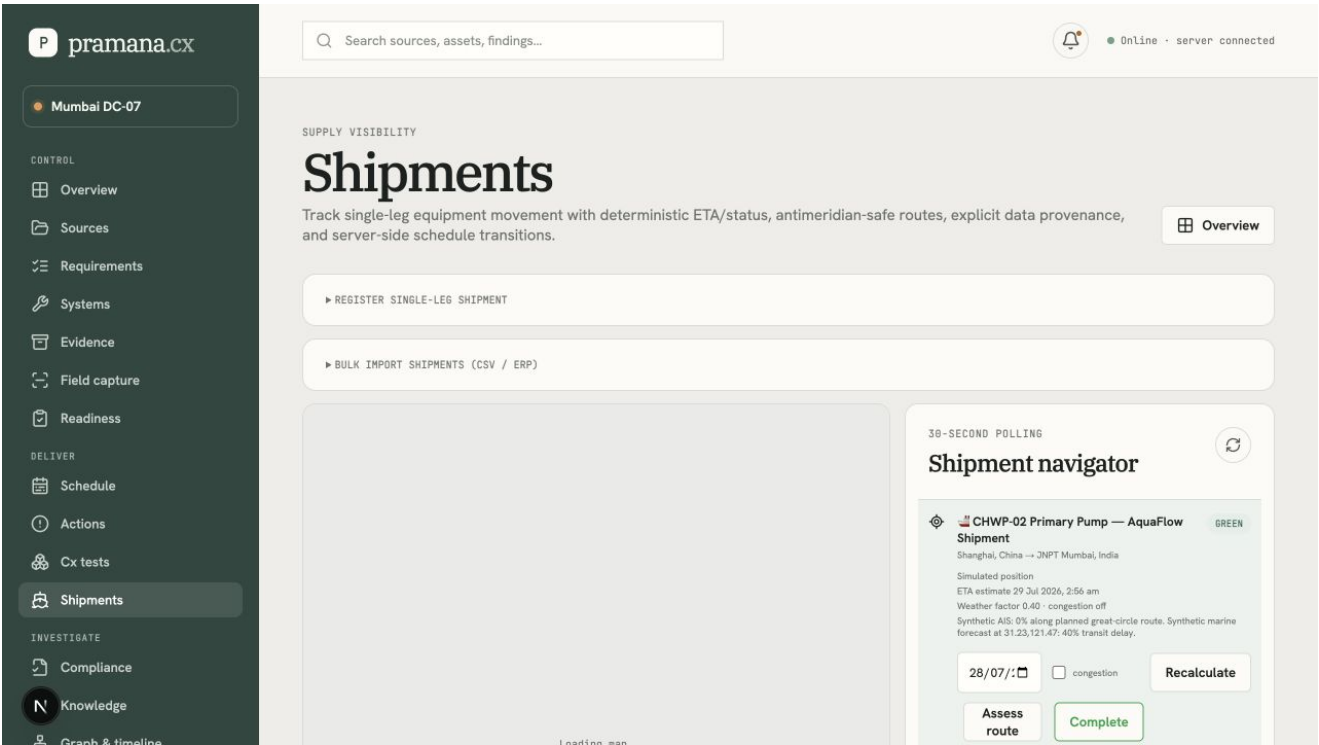
Select equipment

GOVERNING STANDARD VERSIONS

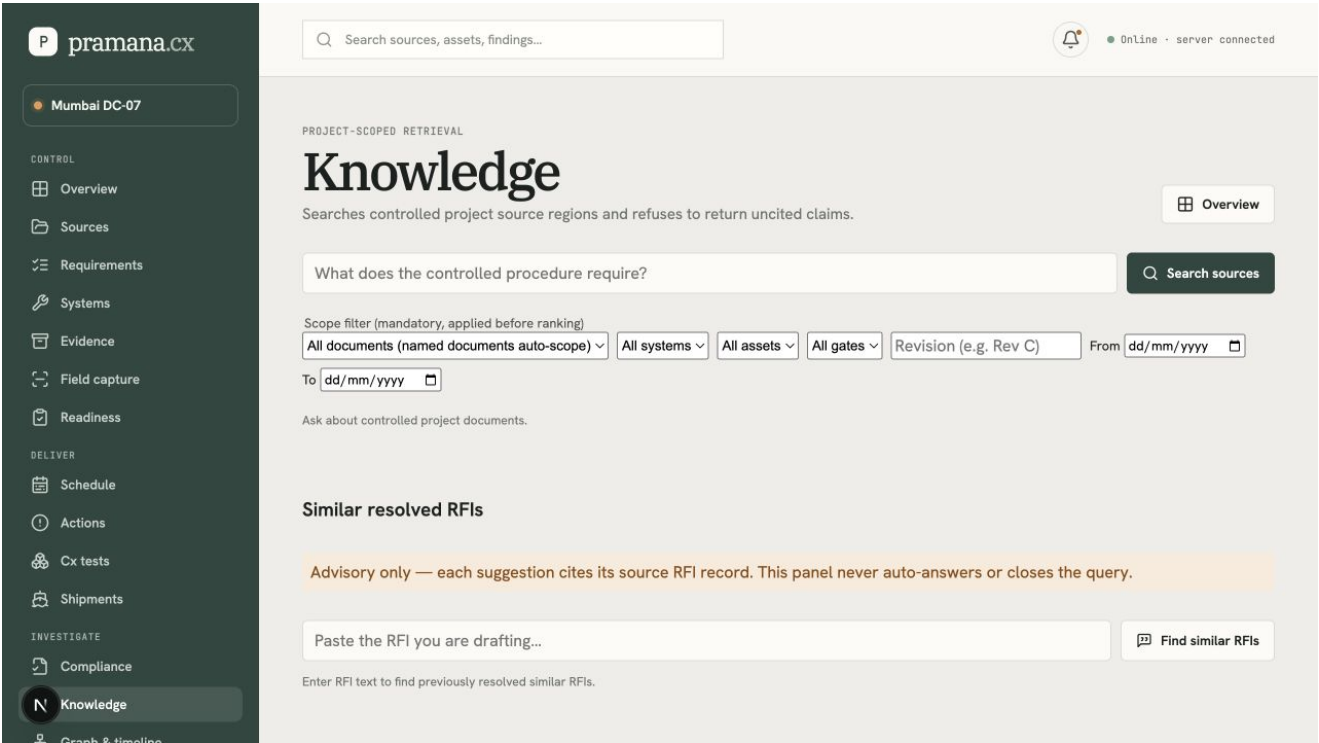
TIA-942 · tia · 1
Unclassified set · CHW Plant Commissioning Procedure · Rev C
ASHRAE · ASHRAE Std 90.1-2022 — Energy Standard · Rev 2022
NFPA · NFPA 2001 — Clean Agent Fire Suppression · Rev 2018
Unclassified set · CHW Plant Commissioning Procedure · Rev C

Generate draft

Governed Cx tests - cited standards, deterministic readings, human review and report approval.



Shipment visibility - route/position context, ETA provenance and controlled operational status.



Cited knowledge intelligence - project-scoped retrieval and source-grounded answers.

End-to-end user journey

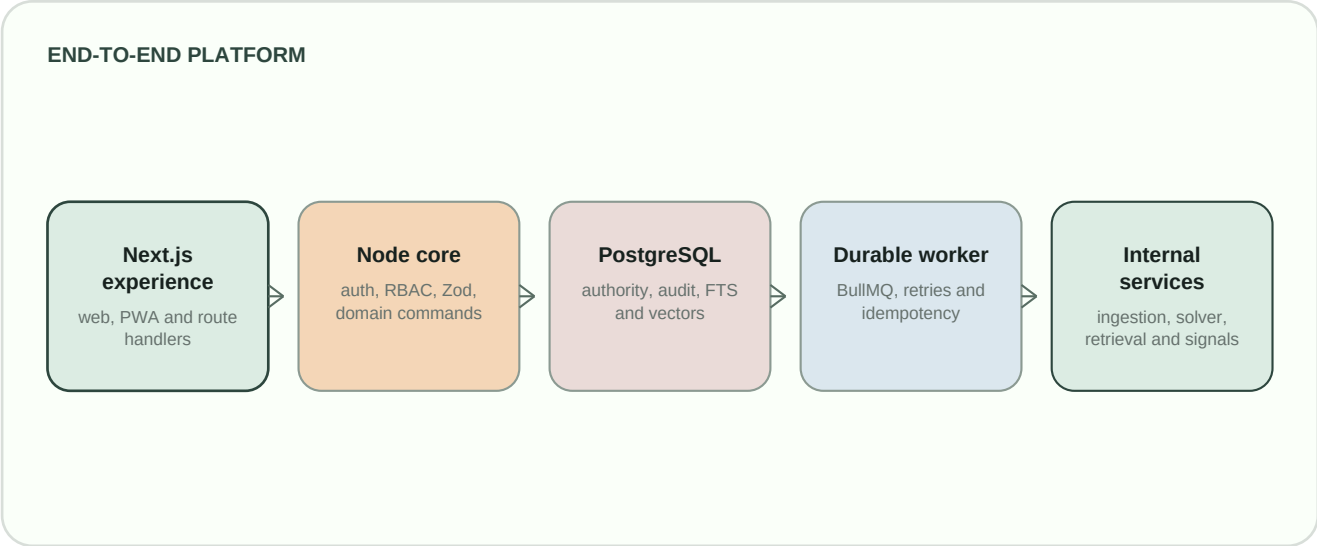
Step	Surface	Data flow	Outcome
1	Overview / Settings	Project membership scopes all loads and actions.	User enters an authorized project context.
2	Sources / Cx	Bytes -> hash/version -> source regions -> durable extraction.	Reviewer sees cited proposals.
3	Systems / Requirements	Accepted records create typed system/asset/gate relationships.	Facility context is traversable.

Step	Surface	Data flow	Outcome
4	Evidence / Field / Cx	Evidence starts pending; review adds acceptance/audit state.	Proof contributes to readiness only when accepted.
5	Schedule / Shipments	Validated event -> delta detector -> CP-SAT only if material.	New immutable version or status-only update.
6	Actions / Compliance / Knowledge	Findings, alerts and citations retain record identity.	Owner resolves human-reviewed work.
7	Readiness / Turnover	Rules + authorized decision -> manifest.	Approved gates produce verifiable turnover.

03 / SYSTEM ARCHITECTURE

Modular monolith with isolated services

The Next.js core is the authenticated, project-scoped front door and business authority. Durable or expensive work passes through Redis/BullMQ. PostgreSQL is the transactional source of truth; object storage holds immutable bytes; typed relational edges provide governed graph traversal.



Service ownership and boundaries

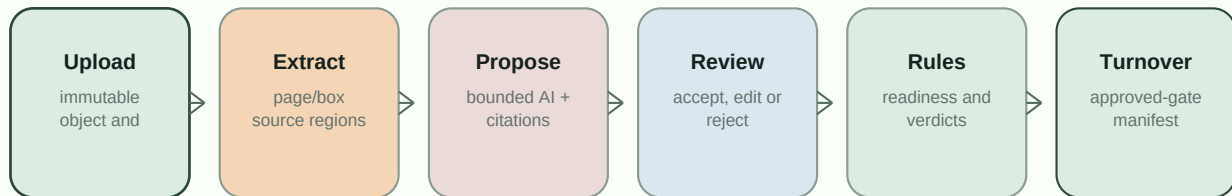
Component	Owns	Must not own
Next.js core	Authentication, authorization, validation, domain commands, audit and project scoping.	Long solver/extraction work or vendor-specific direct AI calls.
Worker + BullMQ	Retryable background operations, stable job identity and durable job state.	Interactive authorization or final human approval.
Ingestion service	Format extraction into structured regions.	Business authority, readiness or permission decisions.
CP-SAT solver	Feasible schedule or explicit bottleneck/infeasibility result.	Database/object storage access, approvals or LLM calls.
Retrieval service	Embeddings and cross-encoder reranking.	Database credentials, tenant decisions or final-answer authority.

Architecture decision. CP-SAT runs as a stateless dedicated service because an optimization run must not block the Node request process. The caller passes the complete DAG/constraints; the solver does not read Postgres or object storage.

04 / DATA FLOW, USER FLOW AND NAVIGATION

From controlled source to approved turnover

AUTHORITY AND EVIDENCE FLOW

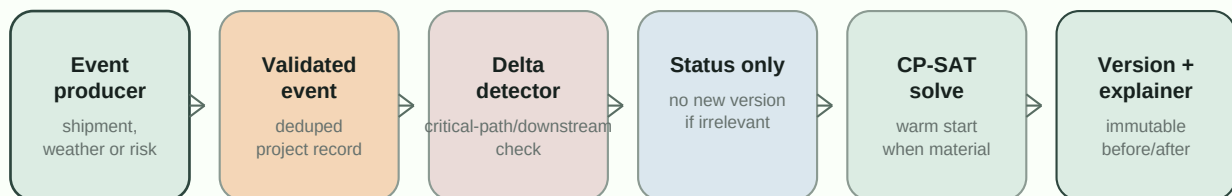


Runtime job lifecycle

- The browser submits a project-scoped command. The core authenticates, authorizes, validates and persists idempotency/audit state.
- Long-running work becomes a stable BullMQ job. The caller receives a job reference and can read progress without freezing the UI.
- The worker reads immutable input, invokes an internal service under timeout/retry rules, validates structured output and persists result, edges and audit event.
- The UI reads the authoritative result and labels advisory versus authoritative state.

Schedule event flow

MATERIALITY-GATED SCHEDULE HANDLING



Schedule safety boundary. Predictive risk can observe, explain and propose mitigation options. It cannot apply a mitigation, alter dependencies/resources, invoke the solver autonomously or change dates.

Navigation contract

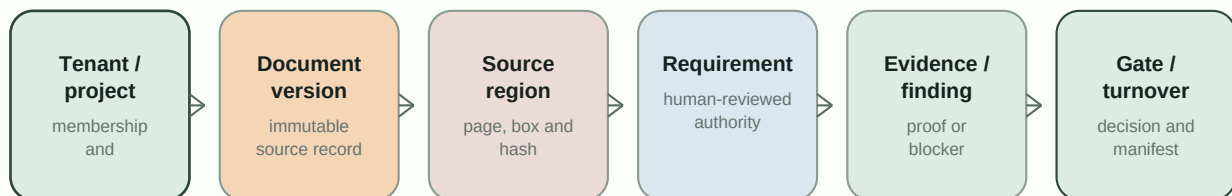
Cross-feature actions include a stable identifier in the URL, for example `/readiness?gate=...`, `/schedule?task=...`, `/actions?finding=...` and `/shipments?shipment=...`. The destination validates, consumes and focuses the exact record; a missing target degrades safely rather than producing an unrelated generic screen.

05 / DATABASE, PROVENANCE AND SCHEMA

Transactional authority, immutable bytes, traversable relationships

Every authoritative record is tenant and project scoped. Cross-project identifiers are rejected at the API boundary. Normalized relational tables remain the authority; the edges relation enables governed traversal; immutable object storage retains original source/evidence/report/turnover bytes.

CORE ENTITY PATH



Persistence domains

Domain	Representative records	Integrity intent
Identity and scope	tenants, users, projects, project_members, sessions	Membership is evaluated server-side for every record and action.
Sources and knowledge	documents, document_versions, source_regions, knowledge_chunks	Immutable bytes, exact citations and embedding-model tags prevent mixed vector spaces.
Evidence control	requirements, evidence, systems, assets, gates, findings, decisions, turnover packs	Human review and deterministic readiness govern state transitions.
Planning and supply	schedule_tasks, resources, schedule_versions, schedule_events, risks, shipments	Accepted inputs, immutable solve history, event deduplication and provenance.
Cross-cutting	edges, durable_jobs, alerts, audit_events	Typed relationships, retry/idempotency and append-only hash-linked history.

Provenance safeguards

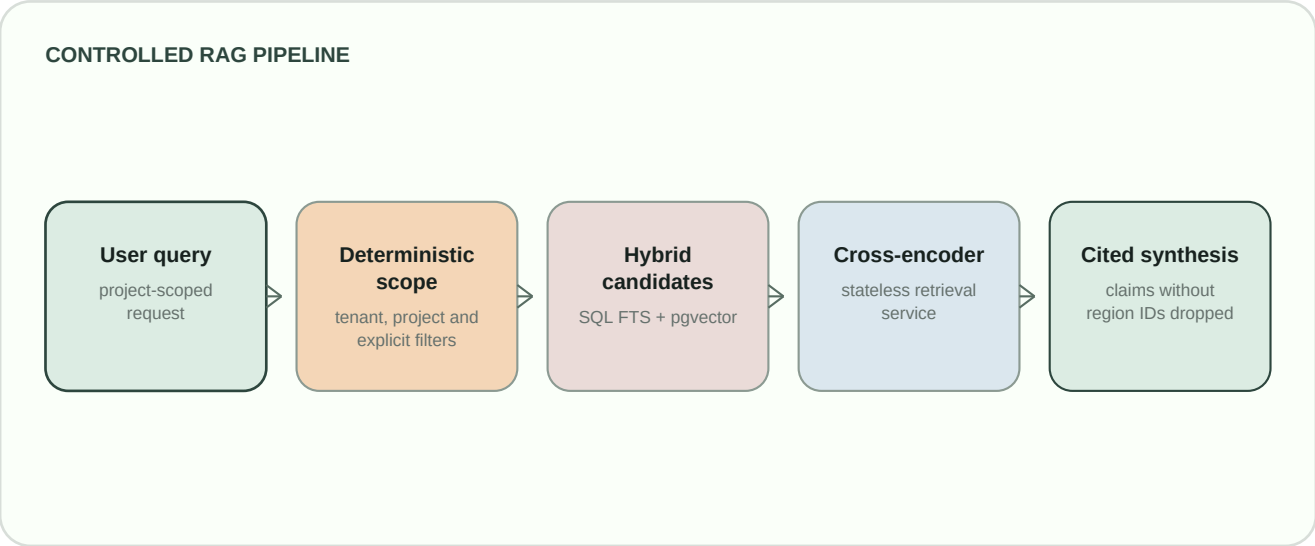
- Immutable SHA-256 object record is stored before extraction; source regions retain document version, page, bounding box and content hash.
- New revisions never overwrite old regions; superseded, rejected, failed or unauthorized sources cannot be retrieval candidates.
- Accepted-only evidence can prove a requirement. Stale/failed evidence invalidates readiness; edits trigger audited recalculation.

- Audit events are append-only and hash-linked per project.

06 / AI, RAG AND AGENT BOUNDARIES

Bounded, cited, time-limited and non-authoritative

Core generation and embedding calls resolve through a central provider boundary. Local Ollama is the intended deployment default: gemma4:e2b for structured generation and nomic-embed-text:latest for 768-dimensional embeddings. Gemini and NVIDIA NIM can be selected explicitly. The deterministic mock provider is verification-only and must never be deployed.



Provider and RAG safeguards

- Bounded prompt/context sizes, output token caps, request deadlines, schema validation and retry limits.
- Client aborts, per-operation busy state and progress feedback prevent frontend blocking.
- Tenant/project and supplied metadata filters are enforced in SQL before ranking, not after retrieval.
- Only explicit reviewer/user filters and deterministic title routing may narrow authority-bearing scope. The LLM cannot choose system, asset, gate, revision, type or document scope.
- Groundedness is code-enforced: an answer claim without a resolvable source-region identifier is dropped.

Five operational agent capabilities

Capability	Implementation posture	Non-negotiable boundary
Commissioning QA Copilot	Cited standards, checklist proposal, deterministic readings, human-routed narrative review and draft report workflow.	No automatic approval; approved report only becomes immutable evidence after review.
Supply Chain Visibility and Risk	Shipment legs, multi-mode map, polling and weather/congestion/AIS assessment.	Delayed/recovered event is provenance-labelled and cannot directly alter a schedule.
Specification and Quality Compliance	Semantic candidate discovery plus deterministic numeric/boolean/category comparisons and precedent handling.	Creates proposed findings only; an engineer decides disposition.
Predictive Schedule Risk	Bounded observations of procurement, lead-time, workforce and weather with materiality/deduplication.	Advisory risks/mitigations only; it cannot change dates or apply an option.
Knowledge and RFI Intelligence	Hybrid project-scoped retrieval, graph context, reranking, cited synthesis and similar-RFI retrieval.	Returns grounded citations or explicit no-results; never invents authority.

Concrete technology inventory

Layer	Technology	Purpose
Web	Next.js 16.2.10, React 19, TypeScript 5.7	Server-rendered application, route handlers, typed UI and responsive PWA surfaces.
Schema and validation	Drizzle ORM 0.45, Drizzle Kit 0.31, Zod 3.24	SQL migrations, relational authority and validated contracts.
Database	PostgreSQL 16 target topology, pgvector, Postgres FTS	Transactional state, constraints, audit/jobs and hybrid search.
Queue/cache	Redis, BullMQ 5.80, ioredis 5.11	Durable background work, idempotency coordination and rate limits.
Storage	Filesystem locally; MinIO/S3 via AWS SDK v3	Immutable original, report and turnover artifacts with signed reads.
Extraction / solver / retrieval	Python/FastAPI, PyMuPDF, Google OR-Tools CP-SAT, BAAI bge models	Controlled source extraction, optimization, embeddings and reranking.
AI providers	Ollama gemma4:e2b and nomic-embed-text; optional Gemini and NVIDIA NIM	Explicit schema-validated advisory provider choices.
Maps and UX	Leaflet, React Leaflet, Turf, searoute-js, OpenStreetMap, Lucide, Three.js	Route visualisation, geospatial geometry and application visual system.
Identity	bcryptjs, OTPAuth, optional Clerk Next.js SDK	Owned credentials/TOTP or external identity adapter with app-owned RBAC.

API group contract

Group	Representative routes	Expectation
Identity	/api/auth/*, /api/profile	Credentials/Clerk, sessions, TOTP and rate limits.
Project authority	/api/projects, /members, /activate, /alerts	Server-side membership and role enforcement.
Evidence control	/sources, /evidence, /requirements, /gates, /turnover-packs	Versioning, citations, review and readiness.
Schedule and risk	/schedule/tasks, /resources, /baseline, /events, /risks	Reviewed input, immutable versions and solver isolation.
Cx/compliance/knowledge	/cx/*, /compliance/*, /knowledge/*, /graph	Cited, scoped, proposed-only advisory flows.

08 / PREREQUISITES, INSTALLATION AND CONFIGURATION

Local topology and production configuration

Local development can use separate PostgreSQL, Redis, ingestion and solver processes or the complete Compose topology. Production must use secret-managed configuration, non-degraded infrastructure and a model endpoint reachable from both web and worker services.

Installation

```
npm install
cp .env.example .env.local
npm run db:migrate
npm run db:seed
npm run system:start

npm run typecheck
npm run build
npm run verify:all

ollama pull gemma4:e2b
ollama pull nomic-embed-text:latest
MODEL_PROVIDER=ollama EMBEDDING_PROVIDER=ollama npm run verify:ollama
```

Environment and infrastructure requirements

Requirement	Local	Production
Runtime	Node 22, npm, Python 3; Docker Desktop for Compose.	Pinned images/runtimes; separate web and worker processes.
Database	Reachable Postgres; migrations applied; pgvector where used.	TLS, backups, restore rehearsal, role separation and empty-db migration proof.
Secrets	.env.local never committed.	Secret manager, rotation policy and no secrets in artifacts/logs.
Object storage	Filesystem or local MinIO.	Private bucket, encryption, lifecycle/retention and signed URL policy.
Models	Optional local Ollama for real smoke tests.	Reachable model service for web/worker with data-processing approval.

Key configuration groups

Group	Representative keys	Release condition
Auth	APP_BASE_URL, AUTH_MODE, AUTH_ENCRYPTION_KEY, Clerk keys	Use credentials or Clerk, never development mode.
AI/RAG	MODEL_PROVIDER, EMBEDDING_PROVIDER, OLLAMA_BASE_URL, model keys, GEMINI_API_KEY, NIM_API_KEY	No mock provider; endpoints reachable and bounded.
Infrastructure	Database/Redis URLs, object storage settings, INFRA_ALLOW_DEGRADED	All dependencies healthy; degraded mode false.
Signals	Procurement, lead-time, workforce, weather, AIS/congestion settings	Live/synthetic/unavailable provenance is explicit.

Installation exit criteria. Migrations pass, dependencies are healthy, an authorized source can be processed, the verification matrix passes and the real-provider smoke test is run separately when a provider is configured.

09 / TESTING, ERROR HISTORY AND MITIGATION

Release confidence is a matrix, not a green build

Area	Coverage	Evidence
Build/migration	TypeScript, optimized build, Drizzle checks and migration replay.	Valid artifact and schema state.
Security/tenancy	Credentials/TOTP, rate limits, cross-project rejection and audit chain.	Server-side enforcement and append-only history.
Evidence to turnover	Pending/accepted proof, readiness, gate decision and manifest verification.	Accepted-only proof and independently verifiable turnover.
Schedule/risk	Inputs, dependencies, CP-SAT feasibility, immutable versions and recovery.	No solve for irrelevant event; explicit degradation.
Cx/compliance/knowledge	Citations, deterministic readings, review, filters, reranking and groundedness.	No ungrounded or unauthorized decision.
Browser acceptance	Sidebar journeys, mobile/desktop layouts, busy state, maps and console capture.	No duplicate-key warning, error boundary or overflow.

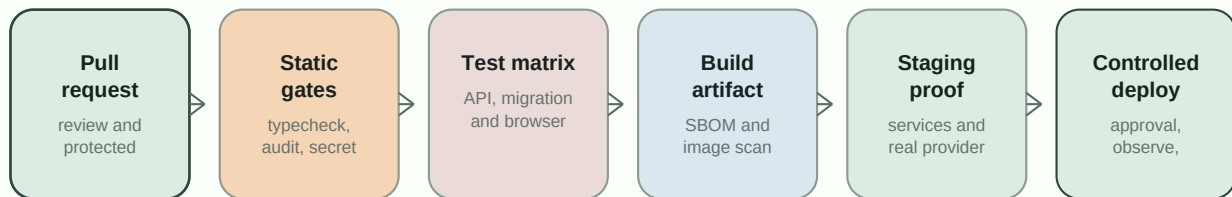
Cross-branch and application mitigation ledger

Failure found	Mitigation now built
Mock TypeScript output could impersonate an LLM.	Explicit generation/embedding providers, Ollama default, mock test-only rule and provenance.
Direct Gemini/provider bypasses produced inconsistent policy.	Central schema-validated provider boundary and real-provider smoke tests.
Long prompts made operations look stuck.	Prompt/token/request bounds, client abort, progress, busy state and bounded retry.
Deep links lost record context.	Destination validation and focus contracts for readiness, schedule, actions and shipments.
Duplicate React keys and business rows.	Stable composite UI keys, unique business constraints and integrity verification.
Conflicting branch migrations.	Final-schema-first reconciliation, ordered migrations and replay validation.
Processed documents were absent/misattributed in RAG.	Extraction-indexing contract, backfill, SQL document filtering and citation validation.
Map fallbacks looked operational.	Coordinate checks, HTTPS/deadline routing, mode-aware provenance and explicit fallback notice.
Raw JSON/hashes/synthetic cues degraded UX.	Presentation helpers, human labels, bounded formatting and source-state labels.
Architecture documentation regressed through merge work.	Visible README diagrams, this PDF dossier, release ledger and documentation-first review.

Local verification result. Type check, optimized production build, production dependency audit and the post-fix local verification matrix have passed. Target-environment infrastructure, real provider acceptance, accessibility and load validation remain production gates.

10 / DEPLOYMENT READINESS AND FUTURE CI/CD

Controlled deployment, not an environment-variable switch

FUTURE CI/CD PIPELINE**Required production gates**

- Secrets and identity: production AUTH_MODE, encryption/session settings, base URL and identity credentials are secret-managed and validated.
- AI: selected Ollama, Gemini or NIM generation/embedding endpoint is reachable by web and worker; deadline, structured output and embeddings are smoke-tested in release configuration.
- Data plane: pgvector is available; migrations-from-empty, S3/MinIO lifecycle, Redis/worker recovery and backup/restore are rehearsed.
- Internal services: ingestion, solver and retrieval health checks are green; TLS/internal network restrictions and non-degraded infrastructure are active.
- Live signals: AIS/weather/congestion/procurement provenance and failure semantics are validated; synthetic paths are disabled or visibly labelled outside test/demo environments.
- Operations: CI, structured logs/correlation IDs, metrics, traces, queue dashboards, alerting, incident ownership, accessibility and load testing are present before release.

Explicit non-claims

Engineering authority remains human. Pramana Cx assists engineering judgement. It does not independently certify a facility, replace the commissioning authority or engineer of record, issue Tier/TIA/BICSI/statutory approval, autonomously close findings/waivers or represent simulated operational feeds as live evidence.

Maintenance references

README.md - implementation map, diagrams, defect audit and configuration. **STATUS.md** - authoritative shipped/open ledger. **PLANNER/PRD.md** and **PLANNER/TRD.md** - product scope and technical contracts. **PLANNER/Schema.md**, **AppFlow.md**, **DesignDecisions.md** and **RetrievalArchitecture.md** - data model, flows, decisions and retrieval safeguards.

APPENDIX A / COMPLETE COMPONENT ARCHITECTURE

Full component and responsibility map

This appendix records the implementation-level architecture omitted by an executive summary. It identifies every major runtime boundary, its state, its caller and the rule that prevents it from becoming an unauthorized decision-maker.

Layer	Components	Responsibility	Hard boundary
Experience	Next.js server-rendered pages; responsive browser shell; mobile navigation; field-capture PWA; Three.js 404.	Project workbenches for Overview, Sources, Requirements, Systems, Evidence, Field Capture, Readiness, Schedule, Actions, Cx, Shipments, Compliance, Knowledge, Graph, Command Center, Changes and Turnover.	UI only requests project-scoped APIs; it does not make authority decisions locally.
Core API	Next.js route handlers; Zod contracts; permission middleware; model-provider facade; presentation helpers.	Authenticates, authorizes, validates, persists domain commands, returns read models and controls provider selection.	No direct unscoped data access or direct vendor-model bypass.
Business authority	Requirements/evidence/gates/finding s/decisions; readiness rules; Cx deterministic verdicts; audit chain; typed edges.	Makes relational state authoritative and invokes deterministic calculations over accepted records.	LLM output has proposed state only; no AI can mark a gate ready.
Durable execution	Redis, BullMQ, durable_jobs, idempotency_records, worker and bounded retry wrappers.	Handles parse, embed, solve, polling, report and other retryable work with stable job identity.	Jobs cannot bypass review/permission checks; retries are bounded and observable.
Ingestion	PyMuPDF service on 8001; PDF/CSV/XLSX parser; source-region extraction; object-storage boundary.	Creates page/box/row/cell provenance and structured extraction output from immutable source versions.	Never becomes a source of business truth without core validation and human review.
Scheduling	FastAPI OR-Tools CP-SAT service on 8002; accepted task/resource inputs; delta detector; immutable versions; explainer.	Solves feasibility/optimization and returns explicit result or bottleneck report; carries completed work as fixed input on warm start.	Solver is stateless, has no database credentials and never changes a date without a reviewed command.
Retrieval	Postgres FTS + pgvector; knowledge_chunks; graph context expansion; retrieval service on 8003; cross-encoder reranker.	Retrieves project-scoped controlled regions and ranks candidates before cited synthesis.	Mandatory tenant/project/metadata filters happen before rank; service has no database credentials.
Operational signals	Shipments, AIS/weather/congestion, procurement/lead-time/workforce, schedule_events, risks and alerts.	Persists observations, calculates materiality/deduplication and exposes advisory impact.	All sources explicitly label live, synthetic or unavailable; no signal directly changes readiness/schedule.
Object and security	PostgreSQL, pgvector, local/MinIO/S3 objects, signed URLs, credentials/TOTP or Clerk, rate limits.	Owns transactional state, immutable artifacts and controlled access.	Cross-project IDs rejected; audit is append-only/hash-linked; production secrets are external.

Full agent event contract

Event	Producer	Trigger	Persisted impact
TEST_FAILED	Cx workflow	Deterministic numeric/threshold or boolean/presence proposed failure only.	Finding/NCR proposal and blocked gate flow; narrative observations stay human review.
shipment_delayed	Shipment assessment	Status transition into at-risk/delayed, never a repeated poll state.	One event per affected task fan-out; delta detector assesses schedule impact.
shipment_recovered	Shipment assessment	Prior delay recovers to on-time.	Clears stale active alert and may enter the same delta assessment.

Event	Producer	Trigger	Persisted impact
predicted_risk_delay	Predictive risk poll	Task + risk-type materiality threshold crosses or changes materially.	Advisory risk/mitigation event; no direct re-solve or schedule mutation.

APPENDIX B / ARCHITECTURE DECISION RECORDS

All committed architecture decisions

ID	Decision	Implementation consequence
ADR-001	Local modular monolith	One typed core for MVP authority; isolated services only for extraction, solve and retrieval.
ADR-002	Postgres + Drizzle	Normalized relational source of truth with migrations and constrained traversal.
ADR-003	Object-store bytes	Immutable originals and exports live behind one local/S3-compatible boundary.
ADR-004	Hybrid lexical + semantic retrieval	Exact FTS plus project-scoped vector candidates and reranking.
ADR-005	Durable asynchronous ingestion	Long work runs through retryable jobs and human checkpoints.
ADR-006	Deterministic readiness	AI can propose; rules and authorized humans decide authority-bearing state.
ADR-007	Project-scoped RBAC + strong approval	Server-side membership with TOTP gate protection.
ADR-008	Offline-capable field PWA	Idempotent local queue supports site capture without authorizing offline decisions.
ADR-009	Dedicated CP-SAT service	Official OR-Tools backend avoids unstable WASM and Node event-loop blocking.
ADR-010	ModelProvider for schedule extraction	Ambiguous extracted tasks/resources require human review.
ADR-011	Deterministic solver objective order	Deadline overrun first, idle time second; never delegated to an LLM.
ADR-012	Immutable schedule versions	No in-place schedule mutation; every solve produces inspectable history.
ADR-013	Delta-gated re-solves	Only critical/downstream impact invokes CP-SAT.
ADR-014	Internal agent-service pattern	Domain services integrate through the same controlled boundary as solver.
ADR-015	Pinned event contract	Event payload/dedup semantics are defined while a general orchestrator remains deferred.
ADR-016	Cx proposed verdict + draft report	Narrative remains human-routed; report must be approved before evidence authority.
ADR-017	Leaflet shipment navigator	Route geometry, click-to-zoom and explicit live/simulated labels.
ADR-018	LLM confined to drafting	Acceptance/shipment classification remain deterministic or human reviewed.
ADR-019	Compliance proposed-findings queue	Modality-tiered flags and clause-vs-line evidence diffs require disposition.
ADR-020	Predictive risk as advisory poll	Dedicated live-events/risk surfaces cross-link critical path but do not reschedule.
ADR-021	Scoped knowledge + edges graph	User-facing cited RAG and graph/timeline reuse project data, no parallel graph authority.
ADR-022	Command Center as read/cross-link surface	Unified deduplicated alerts, recovery clearing and no direct state mutation.

APPENDIX C / COMPLETE DATABASE INVENTORY

Schema domains and real table inventory

The current Drizzle schema defines 45 tables. The following inventory preserves every authoritative domain rather than reducing the database to a generic ER picture.

Domain	Tables	Authority and relationship role
Scope and identity	tenants, users, auth_sessions, projects, project_members	Tenant/project ownership, membership, sessions and RBAC.
Object/source control	storage_objects, documents, document_versions, source_regions	Immutable objects, revision lineage and exact extraction provenance.
Facility and authority	systems, assets, gates, requirements, evidence, findings, decisions	Controlled requirements, proof, blockers and authorized decisions.
Graph and audit	edges, audit_events, durable_jobs, idempotency_records, alerts	Typed traversal, hash chain, job lifecycle, dedupe and alert state.
Commissioning	cx_checklists, cx_checklist_steps, cx_clause_citations, cx_test_records, cx_step_results	Cited checklist execution, deterministic readings and report evidence.
Compliance and knowledge	compliance_precedents, compliance_checks, knowledge_chunks	Proposed deviations, exact precedents, model-tagged hybrid retrieval units.
Logistics/turnover	shipments, turnover_packs	Asset-linked delivery observations and approved manifest artifacts.
Scheduling	schedule_tasks, schedule_resources, schedule_task_resources, schedule_dependencies, schedule_versions, schedule_assignments	Reviewed DAG/capacity inputs, immutable versions and assignments.
Risk and learning	schedule_events, risk_signals, schedule_risks, teachback_notes	Validated event history, source observations, advisory risks and reviewed teach-back.

Relationship rules

- Tenant owns projects; project membership authorizes every project-scoped command and read.
- Document versions own source regions; source regions ground requirements, evidence, Cx citations, knowledge chunks and compliance comparisons.
- Systems contain assets; gates govern requirements; evidence can prove requirements and affect assets; findings block gates.
- Edges represent typed relationships such as PROVES, AFFECTS, REQUIRES and PRECEDES, but normalized tables remain business authority.
- Schedule versions contain assignments; tasks have dependencies/resources and are observed by risk signals/events without folding advisory risk into readiness.
- Business-key uniqueness, exact-row reconciliation and relational integrity checks protect merged/seeded data from duplicate identity corruption.

APPENDIX D / COMPLETE API ENDPOINT CATALOG

All route-handler surfaces

The following catalog is generated from the current `src/app/api/**/route.ts` tree at PDF-build time. Dynamic route segments remain visible so contract owners can identify the entity scope.

Group	Endpoint
login	/apiauth/login
logout	/apiauth/logout
password	/apiauth/password
register	/apiauth/register
sessions	/apiauth/sessions/[sessionId]
totp	/apiauth/totp/challenge
totp	/apiauth/totp/disable
totp	/apiauth/totp/enroll
totp	/apiauth/totp/verify
checks	/apicompliance/checks/[checkId]/review
precedents	/apicompliance/precedents/[precedentId]/review
checklists	/apicx/checklists/[checklistId]/review
checklists	/apicx/checklists/[checklistId]
checklists	/apicx/checklists/[checklistId]/steps/[stepId]/reading
reports	/apicx/reports/[reportId]
test-records	/apicx/test-records/[testRecordId]/report/approve
test-records	/apicx/test-records/[testRecordId]/report
[versionId]	/apidocument-versions/[versionId]/assess-change
[documentId]	/apidocuments/[documentId]/revisions
[evidenceId]	/apievidence/[evidenceId]/review
[findingId]	/apifindings/[findingId]
[gateId]	/apigates/[gateId]/decisions
root	/apihealth
[jobId]	/apijobs/[jobId]
[...key]	/apiobjects/[...key]
root	/apiprofile
[projectId]	/apiprojects/[projectId]/activate
[projectId]	/apiprojects/[projectId]/alerts/[alertId]
[projectId]	/apiprojects/[projectId]/alerts
[projectId]	/apiprojects/[projectId]/assets
[projectId]	/apiprojects/[projectId]/audit/verify
[projectId]	/apiprojects/[projectId]/compliance/checks

Group	Endpoint
[projectId]	/api/projects/[projectId]/compliance/precedents
[projectId]	/api/projects/[projectId]/compliance/scan
[projectId]	/api/projects/[projectId]/cx/checklists/[checklistId]/steps/[stepId]/vision-reading
[projectId]	/api/projects/[projectId]/cx/checklists
[projectId]	/api/projects/[projectId]/cx/standards
[projectId]	/api/projects/[projectId]/edges
[projectId]	/api/projects/[projectId]/entropy
[projectId]	/api/projects/[projectId]/evidence
[projectId]	/api/projects/[projectId]/field-captures
[projectId]	/api/projects/[projectId]/findings
[projectId]	/api/projects/[projectId]/gates/[gateId]/readiness
[projectId]	/api/projects/[projectId]/gates
[projectId]	/api/projects/[projectId]/graph/nodes/[nodeId]
[projectId]	/api/projects/[projectId]/graph
[projectId]	/api/projects/[projectId]/knowledge/query
[projectId]	/api/projects/[projectId]/knowledge/rfi-similar
[projectId]	/api/projects/[projectId]/members/[memberId]
[projectId]	/api/projects/[projectId]/members
[projectId]	/api/projects/[projectId]/objects/[objectId]/url
[projectId]	/api/projects/[projectId]/overview
[projectId]	/api/projects/[projectId]/readiness/recompute
[projectId]	/api/projects/[projectId]
[projectId]	/api/projects/[projectId]/schedule/baseline
[projectId]	/api/projects/[projectId]/schedule/current
[projectId]	/api/projects/[projectId]/schedule/dependencies
[projectId]	/api/projects/[projectId]/schedule/live-events
[projectId]	/api/projects/[projectId]/schedule/resources
[projectId]	/api/projects/[projectId]/schedule/risks
[projectId]	/api/projects/[projectId]/schedule/tasks
[projectId]	/api/projects/[projectId]/schedule/versions
[projectId]	/api/projects/[projectId]/shipments/bulk
[projectId]	/api/projects/[projectId]/shipments
[projectId]	/api/projects/[projectId]/sources
[projectId]	/api/projects/[projectId]/systems
[projectId]	/api/projects/[projectId]/turnover-packs
root	/api/projects
[requirementId]	/api/requirements/[requirementId]/review

Group	Endpoint
events	/apischedule/events
resources	/apischedule/resources/{resourceId}/review
risks	/apischedule/risks/{riskId}/review
risks	/apischedule/risks/recurring
tasks	/apischedule/tasks/{taskId}/review
versions	/apischedule/versions/{versionId}/diff
versions	/apischedule/versions/{versionId}/explanation
versions	/apischedule/versions/{versionId}
{shipmentId}	/apishipments/{shipmentId}
{packId}	/apiturnover-packs/{packId}/verify

APPENDIX E / ENVIRONMENT AND CONFIGURATION INVENTORY

Every example configuration key

The environment appendix is generated from the current `.env.example` file at PDF-build time. Values shown are examples only; production values must be secret-managed and individually validated.

Key	Group	Example/default
DATABASE_URL	data/infrastructure	postgresql://pramana:pramana@localhost:5432/pramana
AUTH_MODE	identity/application	development
AUTH_ENCRYPTION_KEY	identity/application	replace-with-at-least-32-random-characters
SESSION_COOKIE_NAME	identity/application	pramana_session
SESSION_TTL_HOURS	identity/application	24
APP_BASE_URL	identity/application	http://localhost:4173
NEXT_PUBLIC_CLERK_PUBLISHABLE_KEY	runtime	required/optional by selected mode
CLERK_SECRET_KEY	identity/application	required/optional by selected mode
REDIS_URL	data/infrastructure	redis://localhost:6379
REDIS_PREFIX	data/infrastructure	pramana
INFRA_ALLOW_DEGRADED	runtime	true
OBJECT_STORAGE_DRIVER	data/infrastructure	local
S3_ENDPOINT	data/infrastructure	http://localhost:9000
S3_REGION	data/infrastructure	us-east-1
S3_ACCESS_KEY	data/infrastructure	minioadmin
S3_SECRET_KEY	data/infrastructure	minioadmin
S3_BUCKET	data/infrastructure	pramana-local
MODEL_PROVIDER	AI/retrieval	ollama
OLLAMA_BASE_URL	AI/retrieval	http://127.0.0.1:11434
OLLAMA_MODEL	AI/retrieval	gemma4:e2b
OLLAMA_EMBEDDING_MODEL	AI/retrieval	nomic-embed-text:latest
MODEL_TIMEOUT_MS	AI/retrieval	45000
MODEL_PROMPT_MAX_CHARS	AI/retrieval	60000
MODEL_OUTPUT_MAX_TOKENS	AI/retrieval	512
MODEL_CONTEXT_TOKENS	AI/retrieval	8192
GEMINI_API_KEY	AI/retrieval	required/optional by selected mode
GEMINI_MODEL	AI/retrieval	gemini-2.5-flash
GEMINI_EMBEDDING_MODEL	AI/retrieval	text-embedding-004
NIM_BASE_URL	AI/retrieval	https://integrate.api.nvidia.com/v1
NIM_API_KEY	AI/retrieval	required/optional by selected mode
NIM_MODEL	AI/retrieval	meta/llama-3.3-70b-instruct
EMBEDDING_PROVIDER	AI/retrieval	ollama

Key	Group	Example/default
RETRIEVAL_SERVICE_URL	AI/retrieval	http://localhost:8003
KNOWLEDGE_SIMILARITY_THRESHOLD	runtime	0.2
INGESTION_SERVICE_URL	domain/targets	http://localhost:8001
SOLVER_SERVICE_URL	domain/targets	http://localhost:8002
LOCAL_UPLOAD_DIR	data/infrastructure	./data/uploads
SHIPMENT_STATUS_BUFFER_HOURS	domain/targets	72
RISK_POLL_MODE	domain/targets	synthetic
RISK_PROBABILITY_THRESHOLD	domain/targets	0.5
RISK_DELAY_HOURS_THRESHOLD	domain/targets	8
COMPLIANCE_DEVIATION_ACCURACY_TARGET	domain/targets	0.95
RISK_LEAD_TIME_TARGET_HOURS	domain/targets	48
RFI_MATCH_ACCURACY_TARGET	domain/targets	0.85
HIGH_SEVERITY_PRECISION_TARGET	domain/targets	0.9
PACK_PREP_TIME_REDUCTION_TARGET	domain/targets	0.6
SOLVER_TIMEOUT_MS	domain/targets	90000
SOLVER_MAX_ATTEMPTS	domain/targets	3
AUTH_RATE_LIMIT	identity/application	10
AUTH_RATE_LIMIT_WINDOW_SECONDS	identity/application	60
UPLOAD_RATE_LIMIT	runtime	20
UPLOAD_RATE_LIMIT_WINDOW_SECONDS	runtime	60
SCHEDULE_RATE_LIMIT	runtime	30
SCHEDULE_RATE_LIMIT_WINDOW_SECONDS	runtime	60
EXPORT_RATE_LIMIT	runtime	10
EXPORT_RATE_LIMIT_WINDOW_SECONDS	runtime	60
POSTGRES_IMAGE	data/infrastructure	required/optional by selected mode
REDIS_IMAGE	data/infrastructure	required/optional by selected mode
MINIO_IMAGE	data/infrastructure	required/optional by selected mode
OLLAMA_IMAGE	AI/retrieval	required/optional by selected mode
POSTGRES_DB	data/infrastructure	required/optional by selected mode
POSTGRES_USER	data/infrastructure	required/optional by selected mode
POSTGRES_PASSWORD	data/infrastructure	required/optional by selected mode
MINIO_ROOT_USER	data/infrastructure	required/optional by selected mode
MINIO_ROOT_PASSWORD	data/infrastructure	required/optional by selected mode

APPENDIX F / VERIFICATION COMMAND INVENTORY

Every verification script currently declared

The complete test command catalog below is generated from package . json. It makes the release matrix auditable and prevents a single green build from being mistaken for end-to-end verification.

Command	Runner
verify:audit	tsx scripts/verify-audit-chain.ts
verify:phase0	tsx scripts/verify-phase0.ts
verify:credentials-http	tsx scripts/verify-credentials-http.ts
verify:schedule-http	tsx scripts/verify-schedule-http.ts
verify:cx-http	tsx scripts/verify-cx-http.ts
verify:compliance-http	tsx scripts/verify-compliance-http.ts
verify:risk-http	tsx scripts/verify-risk-http.ts
verify:poll-http	tsx scripts/verify-poll-http.ts
verify:risk-autopoll-http	tsx scripts/verify-risk-autopoll-http.ts
verify:supply-poll-http	tsx scripts/verify-supply-poll-http.ts
verify:weather-poll-http	tsx scripts/verify-weather-poll-http.ts
verify:risk-http-clients	tsx scripts/verify-risk-http-clients.ts
verify:knowledge-embed	tsx scripts/verify-knowledge-embed.ts
verify:knowledge-query-http	tsx scripts/verify-knowledge-query-http.ts
verify:command-links-http	tsx scripts/verify-command-links-http.ts
verify:rfi-similar-http	tsx scripts/verify-rfi-similar-http.ts
verify:evidence-turnover-http	tsx scripts/verify-evidence-turnover-http.ts
verify:graph-expansion-http	tsx scripts/verify-graph-expansion-http.ts
verify:live-events-http	tsx scripts/verify-live-events-http.ts
verify:turnover-cx-http	tsx scripts/verify-turnover-cx-http.ts
verify:gate-context-http	tsx scripts/verify-gate-context-http.ts
verify:change-impact-http	tsx scripts/verify-change-impact-http.ts
verify:hardening-http	tsx scripts/verify-hardening-http.ts
verify:polling-http	tsx scripts/verify-polling-http.ts
verify:knowledge-http	tsx scripts/verify-knowledge-http.ts
verify:model-provider	tsx scripts/verify-model-provider.ts
verify:provider-safety	tsx scripts/verify-provider-safety.ts
verify:ollama	tsx scripts/verify-ollama-provider.ts
verify:deep-links	tsx scripts/verify-deep-links.ts
verify:data-integrity	tsx scripts/verify-data-integrity.ts
verify:live-providers	tsx scripts/verify-live-providers-fail-closed.ts
verify:retrieval-service	tsx scripts/verify-retrieval-service.ts

Command	Runner
verify:knowledge-rerank	tsx scripts/verify-knowledge-rerank-http.ts
verify:knowledge-synthesis	tsx scripts/verify-knowledge-synthesis-http.ts
verify:compliance-scan-http	tsx scripts/verify-compliance-scan-http.ts
verify:compliance-llm-http	tsx scripts/verify-compliance-llm-http.ts
verify:risk-mitigations-http	tsx scripts/verify-risk-mitigations-http.ts
verify:ingestion-formats-http	tsx scripts/verify-ingestion-formats-http.ts
verify:compliance-modality-http	tsx scripts/verify-compliance-modality-http.ts
verify:turnover-provenance-http	tsx scripts/verify-turnover-provenance-http.ts
verify:config-targets	tsx scripts/verify-config-targets.ts
verify:solver-resilience	tsx scripts/verify-solver-resilience.ts
verify:rate-limit-coverage	tsx scripts/verify-rate-limit-coverage.ts
verify:compliance-finding-fields	tsx scripts/verify-compliance-finding-fields.ts
verify:overdue-findings	tsx scripts/verify-overdue-findings.ts
verify:turnover-schedule-provenance	tsx scripts/verify-turnover-schedule-provenance.ts
verify:evidence-entropy	tsx scripts/verify-evidence-entropy.ts
verify:teachback-http	tsx scripts/verify-teachback-http.ts
verify:rfi-resolution	tsx scripts/verify-rfi-resolution.ts
verify:knowledge-filters	tsx scripts/verify-knowledge-filters.ts
verify:all	tsx scripts/verify-all.ts

APPENDIX G / FULL TOP-20 DEFECT AND RESOLUTION LEDGER

Complete merged-branch remediation record

#	Defect	Resolution
01	Mock output could look like a real LLM.	Explicit real provider configuration, Ollama default, production mock rejection and provenance.
02	AI paths bypassed provider architecture.	Shared schema-validated provider boundary across generation, vision, compliance, risk and knowledge.
03	Generation and embeddings could select contradictory providers.	Independent explicit provider configuration plus embedding-model tagging and mismatch exclusion.
04	Long model work could freeze the frontend.	Prompt/context/token limits, server deadline, client abort, progress and bounded retry.
05	Deep links opened a generic page.	Query-ID consumption/validation and focused destination record state.
06	Duplicate React keys could omit or duplicate cards.	Authoritative-id dedupe and composite stable UI keys.
07	Duplicate systems/gates/edges/chunks/alerts were possible.	Business-key constraints, seed reconciliation and integrity verifier.
08	Two branch migration sequences conflicted.	Final schema reconciliation, ordered migration history and replay validation.
09	Processed PDFs could be absent from RAG.	Extraction-to-indexing contract and historical semantic-chunk backfill.
10	RAG could attribute another standard to named document.	Document selection, exact title routing and SQL filtering before ranking.
11	LLM planner could narrow authority scope incorrectly.	Only explicit filters/deterministic title resolution may narrow metadata.
12	Fallback text could masquerade as grounded synthesis.	Structured cited synthesis or explicit no-results; unsupported claims dropped.
13	Cx could mix cross-scope gate/system/asset.	Project/system relationship validation and 422 rejection.
14	Raw hashes/JSON/placeholder characters leaked into UI.	Presentation helpers, labels, bounded precision and contextual provenance links.
15	Long operations had inconsistent feedback.	Per-operation busy state, status messages and disabled conflicting controls.
16	UI navigation was crowded and weakly structured.	Task-oriented groups, responsive navigation and contextual actions.
17	Shipment map could lack route/position context.	Saved-coordinate routes, markers, fit bounds and explicit no-coordinate state.
18	Land-routing fallback could mislead.	HTTPS/deadline routing, mode-aware provenance and safe fallback labels.
19	Synthetic signals could look operational.	Live/synthetic/unavailable provenance and fail-closed client behaviour.
20	No trustworthy release gate existed.	Full verification matrix, audit/dependency checks and explicit production prerequisites.

Source of truth

The complete narrative, original branch attribution and verification references remain in README.md under 'Top 20 defects found across Refinement and Updated-Refinement'. This appendix preserves every remedial category in the printable architecture reference.

APPENDIX H / FEATURE AND USER-STORY COVERAGE

Full planned-to-implemented product map

Stories	Capability group	Controlled outcome
US-01 to US-08	Project access, source ingestion, requirement review, readiness, actions, change impact, decisions and turnover	A project-scoped evidence trail from controlled source to approved, verifiable pack.
US-09 to US-12	Baseline schedule, event-triggered rescheduling, explanation and gate schedule context	Accepted task/resource inputs lead to immutable CP-SAT versions and cross-linked context, not automatic readiness changes.
US-13 to US-18	Standards checklist, IST execution, deterministic checks, failures, reports and evidence linkage	Cited Cx workflow keeps numerical/boolean verdicts deterministic and narrative judgement human-routed.
US-19 to US-23	Shipment tracking, weather ETA, risk status, map navigator and schedule events	Asset-linked visibility with explicit provenance and delta-gated schedule influence.
US-24 to US-25	Specification compliance and approved-equal grounding	Proposed deviations are exact-citation grounded and require human disposition.
US-26 to US-27	Predictive risk and live-event surfaces	Advisory material risks cross-link critical path but never autonomously alter a plan.
US-28 to US-31	Knowledge query, similar RFI, project graph/timeline and unified Command Center	Project-scoped cited intelligence, typed graph navigation and deduplicated operational alerts.

Explicit product boundaries

- No general cross-project chatbot, autonomous commissioning authority, independent certification, auto-closure of NCRs/waivers/tests or AI-issued gate state.
- No hidden substitution of synthetic AIS/weather/model output for verified live operations.
- No unlicensed standards/client/vendor content in reusable demos, prompts, embeddings or training material.
- No second authoritative graph/vector store: Postgres/pgvector/edges remain authority; service-local working state cannot override it.